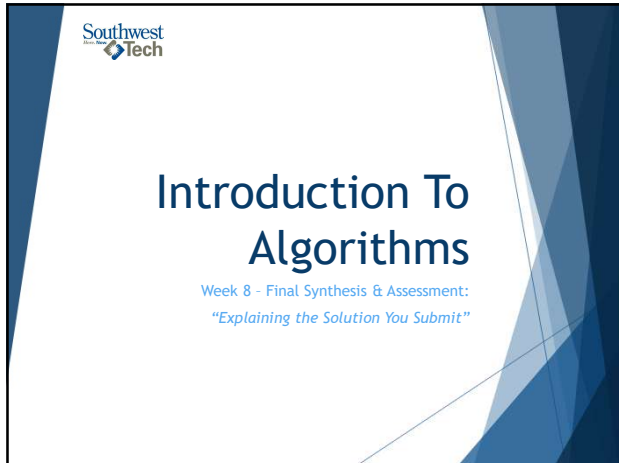
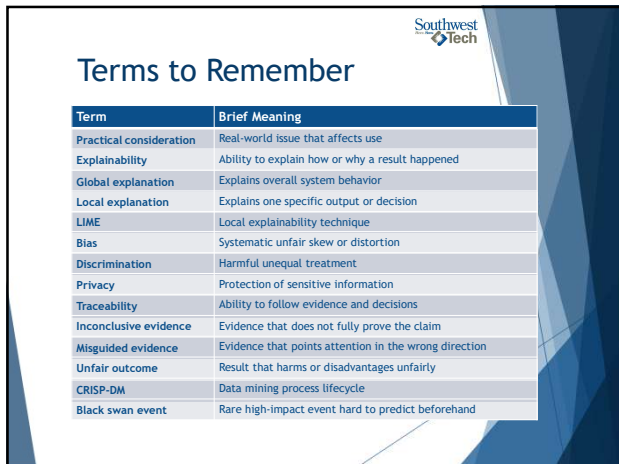




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3

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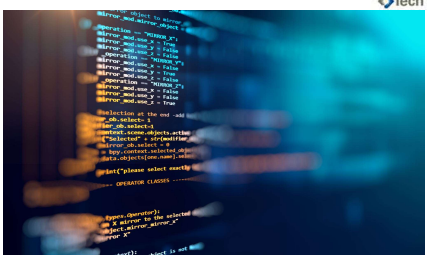
What We Will Explore in Other Courses

- ▶ Formal AI ethics
- ▶ Applied model explainability
- ▶ Larger-scale data validation
- ▶ Security and privacy governance
- ▶ Production monitoring
- ▶ Advanced ML model evaluation
- ▶ Responsible AI system design



4

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Review Coding Lab

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Review: What Lab 07 Taught Us

Last week - A result depended on choices:

- data representation
- similarity rule
- ranking or grouping method
- assumptions
- limitations
- AI/data connection

6

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Today's Question

After an algorithm gives an answer, what still needs to be explained?

7

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Examine the Output with Good Questions

Evidence: What supports this result?

Assumptions: What did we assume?

Limits: Where might this fail?

Impact: Who or what is affected?

Algorithm Output
Result: Recommended Option A
Score: 86

Image Slide

8

Final Week is Synthesis

No Lab!

It is:

- ▶ final synthesis
- ▶ practical-risk discussion
- ▶ explanation practice
- ▶ final assessment preparation & exam

9

What Counts as Success Today

- explain evidence
- name assumptions
- describe tradeoffs
- identify limitations
- discuss responsible AI/tool use
- prepare for the final explanation defense

10

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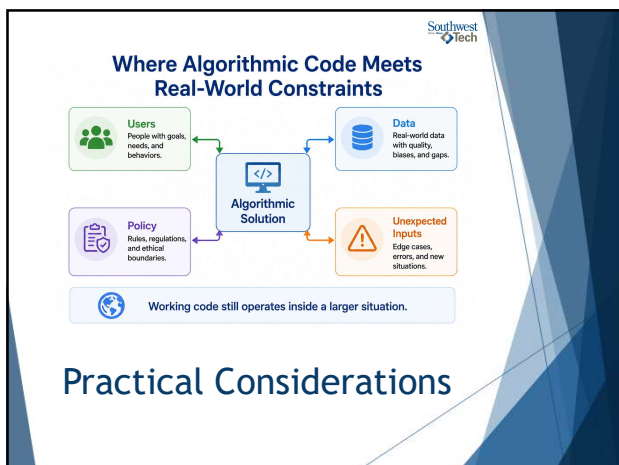
Practical Considerations

Chapter 16 asks:
"What happens when algorithms meet real-world use?"

The reading covers:

- ▶ - unexpected behavior
- ▶ - explainability
- ▶ - ethics
- ▶ - bias
- ▶ - privacy
- ▶ - when algorithms fit
- ▶ - limits of prediction

11



12

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Expecting The Unexpected

Algorithmic solutions face challenges:

- ▶ unusual inputs
- ▶ missing data
- ▶ changing behavior
- ▶ misuse
- ▶ unclear goals
- ▶ real-world consequences

13

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Cautionary Example: Tay

Tay shows that systems can fail when:

- ▶ public input is uncontrolled
- ▶ learning behavior is not bounded
- ▶ safeguards are weak
- ▶ harmful patterns are amplified

14

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Explainability

Explainability asks:

- ▶ Why did the system produce this output?
- ▶ What evidence supports it?
- ▶ What inputs mattered?
- ▶ What assumptions shaped it?
- ▶ What should we not conclude?

15

Algorithm

Text + Text
Text + Graph
Text + Table

ALGORITHMS
Design and Analysis

DATA STRUCTURES
& ALGORITHMS

DISCRETE MATHEMATICS

COMPUTER SCIENCE
FUNDAMENTALS

PROBLEM SOLVING
WITH ALGORITHMS

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Global And Local Strategies

Global explanation:
► explains overall behavior

Local explanation:
► explains one specific output or decision

Both can matter.

16

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Global vs Local Explanation

Global View
How the system usually behaves

Inputs: User Data, Usage Data, Context Data, History Data

Rules: Rule 1, Rule 2, Rule 3, ...

Outputs: Option A, Option B, Option C, ...

How the system usually behaves

Local View
Why did this one result happen?

Input (this case): User: Alex

Rule Applied: If usage is high and engagement is strong, promote Option A.

Output (this result): Recommended Option A

Why this one result happened

Global and Local Strategies

17

Algorithm

Text + Text
Text + Graph
Text + Table

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LIME Recognition

Local Interpretable Model-Agnostic Explanations (LIME)

It is a local explainability technique.

For this course:

- recognize the idea
- do not implement it
- connect it to explaining one result

18

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Ethics And Algorithms

Ethical questions include:

- ▶ Bias (& Typicality Bias)
- ▶ Discrimination
- ▶ Privacy
- ▶ Unfair outcomes
- ▶ Misleading evidence
- ▶ Responsibility for tool use

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When Should We Use Algorithms?

Consider:

- ▶ cost
- ▶ time
- ▶ accuracy
- ▶ risk
- ▶ explainability
- ▶ human impact

Not every problem needs an algorithmic answer.

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Explainability

21

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Working Code Is Not The Whole Answer

Working code is valuable evidence.

But we still ask:

- What problem did it solve?
- What cases were tested?
- What did it assume?
- What could fail?
- What tradeoff did it make?



22

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Global Explanation

Global explanation describes the overall approach.

Example:

"This solution checks eligibility by evaluating account status, training, equipment availability, and overdue-device status before returning a decision."



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Local Explanation

Local explanation describes one output.

Example:

"This request returned 'needs review' because training was complete, but the device type requires supervisor approval."



24

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Evidence Supports Explanation

Useful evidence may include:

- ▶ test table
- ▶ trace table
- ▶ timing table
- ▶ comparison table
- ▶ ranking table
- ▶ README explanation



25

Traceability

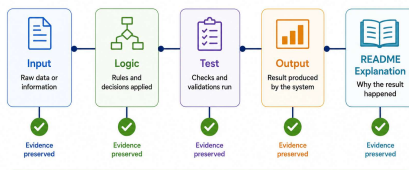
Traceability means a reviewer can follow:

- ▶ input
- ▶ logic
- ▶ evidence
- ▶ output
- ▶ explanation

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Traceability: From Input to Explanation



```
graph LR; Input[Input  
Raw data or information] --> Logic[Logic  
Rules and decisions applied]; Logic --> Test[Test  
Checks and validations run]; Test --> Output[Output  
Result produced by the system]; Output --> README[README  
Explanation  
Why the result happened];
```

Traceability connects what happened to how we explain it.

Traceability

27



Inconclusive Evidence

Inconclusive evidence does not fully prove the claim.

Examples:

- ▶ too few tests
- ▶ missing edge cases
- ▶ tiny dataset
- ▶ only happy-path examples
- ▶ unsupported performance claims

28



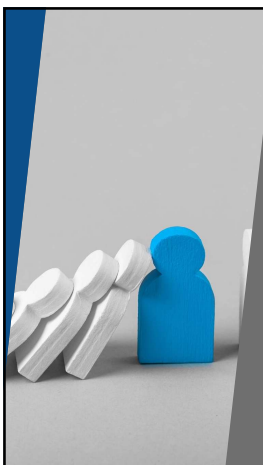
Misguided Evidence

Misguided evidence can look convincing while measuring the wrong thing.

Ask:

- ▶ Does this evidence match the claim?
- ▶ Is the test realistic?
- ▶ Is the dataset representative?
- ▶ Is the conclusion too broad?

29



Limitations Are Not Failure

A limitation is an honest boundary.

It can explain:

- ▶ Where the solution may fail
- ▶ What data is missing
- ▶ What assumption may not hold
- ▶ What should be improved later

30

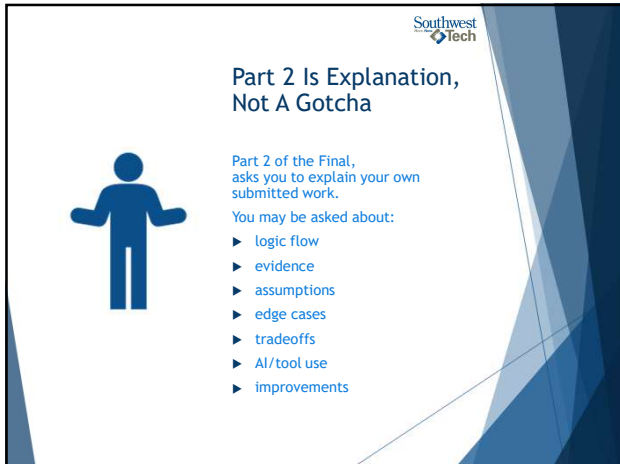


Tradeoff Explanation

Tradeoffs may involve:

- correctness
- readability
- speed
- memory
- simplicity
- fairness
- explainability

31



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Part 2 Is Explanation, Not A Gotcha

Part 2 of the Final,
asks you to explain your own
submitted work.

You may be asked about:

- ▶ logic flow
- ▶ evidence
- ▶ assumptions
- ▶ edge cases
- ▶ tradeoffs
- ▶ AI/tool use
- ▶ improvements

32



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Ethics and Evidence Quality

33

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Bias And Discrimination

Bias can shape algorithmic results.

Watch for:

- ▶ incomplete data
- ▶ skewed examples
- ▶ unfair criteria
- ▶ historical patterns
- ▶ hidden assumptions

34

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How Hidden Data Skew Affects Decisions

Training Data
What the model learns from

- Missing or uneven data
- Some groups overrepresented
- Some groups underrepresented
- Important cases may be missing

Model / Decision Process
Weights evidence to make a decision

Evidence from data
Decision factors

Decision Result
What the system outputs

Outcome may be biased by the data imbalance

⚠ Uneven data can produce uneven outcomes. Better data leads to fairer, more reliable, and more trustworthy decisions.

Image Slide

35

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Privacy

Privacy asks:

- ▶ What data is collected?
- ▶ Is it necessary?
- ▶ Who can see it?
- ▶ How is it protected?
- ▶ Could the output reveal sensitive information?

36



Problems With Learning Algorithms

Learning algorithms may:

- ▶ learn from poor data
- ▶ amplify harmful patterns
- ▶ drift over time
- ▶ behave differently after deployment
- ▶ become difficult to explain

37



Ethical Questions By Context

Ask different questions for:

- ▶ classification
- ▶ regression
- ▶ recommendation
- ▶ data mining

Same principle:
What could go wrong, and who is affected?

38



Factors Affecting Algorithmic Solutions

Results can be affected by:

- ▶ data quality
- ▶ feature choice
- ▶ assumptions
- ▶ model or algorithm choice
- ▶ evaluation method
- ▶ deployment context
- ▶ human interpretation

39



Unfair Outcomes

An unfair outcome may come from:

- ▶ unfair criteria
- ▶ missing context
- ▶ biased data
- ▶ poor thresholds
- ▶ over-automation
- ▶ weak review process

40



Traceability As Protection

Traceability helps answer:

- ▶ What happened?
- ▶ Why did it happen?
- ▶ What evidence supports it?
- ▶ Who reviewed it?
- ▶ What should change?

41



AI Use Accountability

If AI helped, explain:

- ▶ What it helped with
- ▶ What you changed
- ▶ What you tested
- ▶ What you rejected
- ▶ What you still understand and own

42



43

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Previous Work As Reference

You may reference your own previous work.
Be ready to explain:

- ▶ What you reused
- ▶ What you changed
- ▶ Why it fits this task
- ▶ How you verified it

44

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Reducing Bias Is A Process

Bias reduction is not one switch.
It may involve:

- ▶ better problem framing
- ▶ better data review
- ▶ better evaluation
- ▶ more traceability
- ▶ human review
- ▶ revision over time

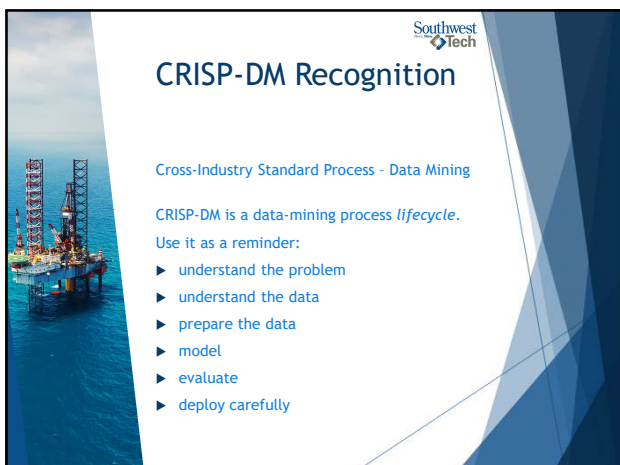
45



46



47



48

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Cost, Time, Accuracy



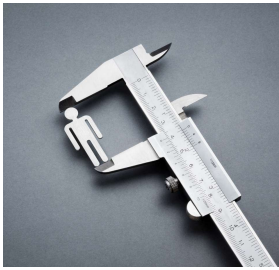
When deciding whether to use an algorithm, ask:

- What does it cost?
- How much time does it save?
- How accurate must it be?
- What happens when it is wrong?

49

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Accuracy Is Not Enough



A solution may be accurate but still problematic if it is:

- ▶ impossible to explain
- ▶ unfair
- ▶ too expensive
- ▶ too slow
- ▶ too brittle
- ▶ harmful when wrong

50

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Black Swan Events

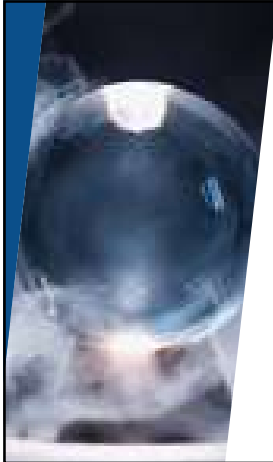


Black swan events are:

- ▶ unexpected
- ▶ high impact
- ▶ easier to explain after they happen
- ▶ not always surprising to everyone

They expose prediction limits.

51



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Forecasting Dilemmas

Hard questions:

- ▶ Can this be predicted?
- ▶ Can the implications be predicted?
- ▶ What data would be needed?
- ▶ What uncertainty remains?
- ▶ Who must decide anyway?

52



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Post-Event Predictability

After an event, people may say:

"It was obvious."

But before the event:

- ▶ evidence may be incomplete
- ▶ signals may be noisy
- ▶ implications may be unclear

53



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Practical Application: COVID-19

COVID-19 illustrates practical limits:

- ▶ changing conditions
- ▶ incomplete data
- ▶ uncertain implications
- ▶ high-stakes decisions
- ▶ need for revision over time

54



Be Careful With Claims

Strong final explanations avoid overclaiming.

Say:

- ▶ "My evidence shows..."
- ▶ "This assumes..."
- ▶ "This may fail when..."
- ▶ "A limitation is..."
- ▶ "With more time, I would..."

55



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Final Bridge & Optional Bonus

56



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Chapter 16 As Final Support

Chapter 16 can help you strengthen final explanations.

Use it for:

- ▶ explainability
- ▶ assumptions
- ▶ limitations
- ▶ bias awareness
- ▶ privacy awareness
- ▶ evidence quality
- ▶ tradeoff reasoning

57

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Optional Enrichment / Bonus Credit

You may be able to earn bonus credit by clearly applying a Chapter 16 concept to your final work.
(Instructors Discretion!!)

Examples:

- ▶ bias risk
- ▶ privacy risk
- ▶ inconclusive evidence
- ▶ black swan limitation
- ▶ cost/time/accuracy tradeoff

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Optional Bonus Credit as an Evidence-Based Add-On

Final Submission

- ✓ Required components completed
- ✓ Meets assignment criteria
- ✓ Ready for evaluation

+

Optional Bonus
Explainability or ethics connection

example

justification

Bonus requires clear evidence and explanation.

Optional Enrichment / Bonus Credit

59

Bonus Credit Must Be Demonstrated

Bonus-worthy application should:

- ▶ connect to your actual solution
- ▶ use evidence
- ▶ explain the risk or tradeoff
- ▶ avoid vague name-dropping
- ▶ appear in README/Part 2 explanation

60



FINAL Exam Structure



The final has two parts:

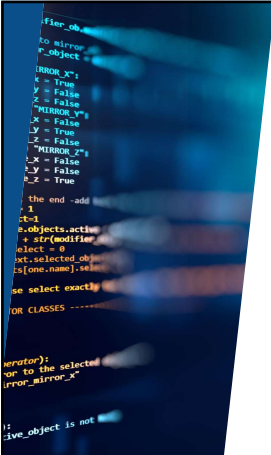
- ▶ Part 1 - Applied Solution Set
- ▶ Part 2 - Explanation Defense

Part 1 is what you submit.
Part 2 is how you explain it

61



Part 1 Readiness



Part 1 should include:

- ▶ working solutions
- ▶ tests and evidence
- ▶ assumptions
- ▶ limitations
- ▶ comparison where required
- ▶ AI/reference-use note when applicable

62



Part 2 Readiness



Be ready to explain:

- ▶ what your code does
- ▶ why a test passed or failed
- ▶ why you chose an approach
- ▶ what assumption matters
- ▶ what limitation remains
- ▶ what AI or previous work contributed

63

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README As Explanation Prep

Your README helps you prepare Part 2.



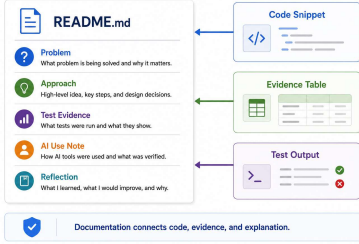
It should organize:

- ▶ task summaries
- ▶ evidence tables
- ▶ assumptions
- ▶ limitations
- ▶ tradeoffs
- ▶ AI/tool-use notes
- ▶ previous-work notes

64

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A Professional README Document, Prepared with Evidence



README.md

- Problem**: What problem is being solved and why it matters.
- Approach**: High-level idea, key steps, and design decisions.
- Test Evidence**: What tests were run and what they show.
- AI Use Note**: How AI tools were used and what was verified.
- Reflection**: What I learned, what I would improve, and why.

Code Snippet

Evidence Table

Test Output

Documentation connects code, evidence, and explanation.

README Prep

65

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Final Explanation Pattern



Use this pattern:

1. What did I build?
2. What evidence shows it works?
3. What did I assume?
4. What tradeoff did I make?
5. What limitation remains?
6. What did I verify?

66



67



68



69
